

EXAM 7 – SPRING 2011

1. (3 points)

Given the following information (\$000) for a line of business:

Accident	Written	Earned	<u>Cumulative Reported Losses</u>		
<u>Year</u>	<u>Premium</u>	<u>Premium</u>	<u>12 Months</u>	<u>24 Months</u>	<u>36 Months</u>
2007	5,756	4,779	413	2,310	5,845
2008	6,907	5,735	0	541	1,309
2009	8,289	6,882	936	2,311	
2010	9,946	8,258	50		

- The tail factor from 36 months to ultimate is 1.050.

a. (1 point)

Use the least squares method to estimate ultimate losses for the 2009 accident year.

b. (0.5 point)

Discuss the reasonability of the estimate derived in part a. above, relative to the estimate that would be produced by the link ratio method.

c. (1.5 points)

Illustrate graphically the relationships between the link ratio method, budgeted loss method and least squares method in modeling the loss development process.

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Sample Solutions
Exam 07, 2011, Question 1

Solution 1

a)

Accident Year	Loss Ratio			
	12 months	24 months	36 months	ult
2007	0.086	2310/4779=0.483	5845/4779=1.223	1.223×1.05=1.284
2008	0.000	541/5735=0.094	1309/5735=0.228	0.228×1.05=0.240
2009	0.136	0.336		
2010	0.006			

$$\bar{x}=0.289$$

$$\bar{y}=0.762$$

$$\overline{x^2}=0.121$$

$$\overline{xy}=0.322$$

$$b = \frac{\overline{xy} - \bar{x} \cdot \bar{y}}{\overline{x^2} - \bar{x}^2} = 2.685$$

$$a = \bar{y} - b\bar{x} = -0.013$$

$$L(x) = -0.013 + 2.68 \times (2311/6882) = 0.89$$

$$\text{Ult loss} = 0.89 \times 6882 = 6111$$

b) In part a, a is less than zero. So our estimate of y will be negative for small value of x. That does not make sense. One might substitute the link ratio method while $a < 0$. Link ratio will produce positive number even for small values of x.

c)

